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Development of Leftover food management system using efficient hunger search techniques

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Abstract— it is no secret that food waste is a big problem. While food waste presents both moral and economic dilemmas, it also creates environmental problems. Food waste has noticeable impacts on hunger and malnutrition, it also effects environment in a severe way. The current improvements to overcome food waste are mainly focused on redirecting the large-scale food leftovers but not focused on house hold food leftovers. Hence, we propose a digital application to channelize the excess food in both large scale and small scale to people in need. Using mobile application as a platform helps us reach larger section of people making it easily scalable. The users who are willing to donate food can register themselves as donors and users who are in need of help can register as receivers. This approach of combining technology and focus on house hold leftovers help us prevent unnecessary food waste. The distribution of the donations will follow the discussed Hunger search Algorithm.

Keywords— Food Waste Management, food leftover management, Mobile application, House hold food leftovers, Haversine formula, elliptic boundary, circular boundary, donation, donor, receiver, helper, volunteer

I. INTRODUCTION

Many people across the globe die of hunger every year and still huge amount of food gets wasted. According to the recent Food Waste Index Report by United Nations Environment Programme [1], in Indian households 50 kg of food is being thrown away per person every year. This excess food usually ends up in landfills, creating potent greenhouse gases which have dreadful environmental implications. According to a figure estimated by the Food and Agricultural Organization (FAO), nearly 40 per cent [7] of the food produced in India is wasted every year due to existing food systems and inefficient supply chains. This loss occurs even before the food reaches the consumer. There is also a significant amount of food waste generated in normal households. Most of the people do not realize how much food they throw away every day, from uneaten leftovers to spoiled produce. By managing food sustainably and reducing waste, we can help businesses and consumers save money, connect communities to help each other, mainly to those who do not have enough to eat, and conserve resources for future generations.

Food waste refers to intentional discards of food fit for consumption. It typically takes place at the retail and consumption stages in the food supply chain through the behavior of organizations and individuals. [2] Considering the amount of people starving every day, the food wastage is undeniable.

Many organizations have taken necessary actions to tackle the food waste crisis, launching initiatives and approaches. As a result, public and private institutions have developed policy, interventions and incentives to encourage consumers to reduce their domestic food waste. [4] The donation of food which is still edible can be seen as a specific application of charity as food is recovered for its original purpose, human intake. [3]

The advent of digital tools has bestowed a different outlook on the food industry. [13] Consumers today have the privilege to choose from a variety of cuisines, anywhere, anytime from a range of food providers listed in the e-commerce space. We believe such a smart digital platform is essential to bridge the gap between donors and respective receivers, where donors can communicate about the excess food and eligible receivers can avail this help to fulfil their needs.

There are a number of digital applications to collect monetary donations from interested donors and use them to contribute to those in need. The main factor most of these platforms do not focus is food-leftovers, unspoiled food which is fit for consumption i.e., house-hold food leftovers contributes the majority share in total wasted food.

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This paper deliberates about a mobile application which can be used to direct the food leftover in both large scale and small scale to those who are actually in need. The users can register themselves as donors and receivers. The donors can post information about the excess food they have which reaches all the eligible receivers through the application. The application runs on effective algorithm which manages to shortlist receivers based on their location and need of food. After this sorting and notifying, the receivers can show their interest in receiving food wherein either donor or receiver can afford themselves for transportation. If both of them cannot transport, another user who is registered as volunteer can help them out. This approach of projecting the leftover food to those who are actually in need can make a big difference in food being wasted.

The authors of this report acknowledge that food donations alone cannot solve the pressing challenge of food waste, as this would require addressing the underlying understanding of value of food is its root cause. However, It is believed that preventing food waste is any human's responsibility and hopefully setting up such platforms can encourage people to actively take part in donations.

II. LITERATURE SURVEY

According to Serhat Murat Alagoz & Haluk Hekimoglu (2012) [11], the food industry is showing a steady growth. According to Sheryl E. Kimes (2011) [12], his study found that control and convenience associated with the online food ordering services were important for users. In our use case even though there are several people offering donations, there is no proper channel to deliver this information to the concerned audience. These food ordering services not only provide us with ease and convenience but also motivate us to build and utilize such platforms to spread information about the donations.

Based on S-O-R model [14], terms such as "mobile push notifications" or "push messages" can be regarded as triggers that influence the consumer as an organism and provoke a reaction, as these notifications can potentially influence the responsive behavior of customers [15].

However, there are certain constraints in reaching every house with extra food and technology was one among them. But considering today's technological developments, it shouldn't be a problem in reaching larger section of people. Below are some of the technical solutions that can help us reach out target audience in an efficient way. These notifications are displayed on the lock screen or in the notification bar of the smartphone and are often called "push notifications".

- *Point to point messaging*

In this architecture, when two applications need to communicate with one another, a connection is established and a connector enables application-1 to speak to application-2 by sending a message. A connector is a bridge that allows the exchange of data or application functionalities in a structured manner. Connectors handle the complexities of message translation, integration and any other relating messages operations used to access the application features. And if it is needed to change either of the applications or add a new one to the network, the interfaces of all the applications have to update as well.

- *Observer Design Pattern*

This mainly consists of two entities, the Object (subject) and the Observers. So basically the Object will maintain a list of its observers, and whenever there is any change in the state or when an event occurs the Object will notify all the observers about the event or state change if any.

- *Publish-Subscribe pattern*

This involves two main entities namely publishers and subscribers. Publishers are the ones who sends the message and the subscribers are basically receivers. Here the subscribers and the publishers are unaware of each other. This is how it basically works, whenever there is a new message, the publisher classifies the message into a certain topic and post to that topic. All the subscribers will show their interest in topics by registering themselves as subscribers of that topic. Whenever a publisher posts to any topic, all the receivers subscribed to that topic will be get the message.

- *Data Flow Architecture*

In this architecture, the whole system is seen as series of transformations applied to data through a series of computational components called filters. This includes mainly two components, pipes and filters. Pipes transmit the data between filters, and filters take input of certain format and do computation on that input and then forwards the transformed input to the next filter until it gives the desired output. The main objective of this pattern is to achieve reusability and modularity. Based on the execution sequence, this architecture is of three types:

1. Batch Sequential
2. Pipe and Filter (Non-sequential)
3. Process Control

Pipe and Filter pattern

This approach emphasizes on incremental transformation of data by successive components. In this approach, the flow of data is driven by data and the whole system is decomposed into components of data source, filters, pipes, and data sinks. The main feature of this architecture is its concurrent and incremented execution.

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III. Related Work

There are few non-profitable organizations and several initiatives and spread across the country volunteering these services which shall be discussed below.

- *Annakshetra*

It is a Jaipur based organization[10] and was the first to start this donate food trend in 2010. They work mainly for daily wage workers, sanitation workers, people located in vegetable markets, and malnourished children, etc. They generally collect foods at night and they distribute to the needy only after they get thumbs up from their medical experts

- *No Food Waste*

No Food Waste is an organization which collects excess food from weddings, parties, events and distribute to those who are hungry. The idea took birth in Coimbatore, where it began with a volunteer and 2 Shopper Bags to collect the surplus food and deliver to the homeless through Public Transportation.

- *Feeding India by Zomato*

This provides the right means for compassionate citizens to begin and manage initiatives to help hungry people locally. Nearly 26,000 plus volunteers across 100 plus cities in India work towards the joint mission of 'better food for more people' and 'zero hunger' to take the dream of ending hunger a step closer to reality.

- *Feeding India*

A non-profit social enterprise aims to solve the problem of hunger and malnutrition in the country. They distribute excess food from restaurants, events, corporates, weddings, etc. to people in need and also run sustained feeding programs through their own kitchens. Feeding India operates with a network of 3,500 plus volunteers in 45 plus cities of India [10]. It has served over 5 million meals to beneficiaries in need.

- *Rise against hunger India*

An international hunger relief organization that distributes food and provides needed help to the world's most vulnerable by channelizing necessary resources to end hunger by 2030.

- *Robin Hood Army*

A non-government-based organization (NGO) located in 41 cities of India. It is a volunteer-based organization who collect food from restaurants and even Convention halls and distribute it to the needy people.

- *Wrap It Don't Waste Food*

A Chennai based organization that collect foods from restaurants, parties and weddings. They provide food to the underprivileged people. It is an initiative associated with "Reunite to Redefine India" [10] awareness program.

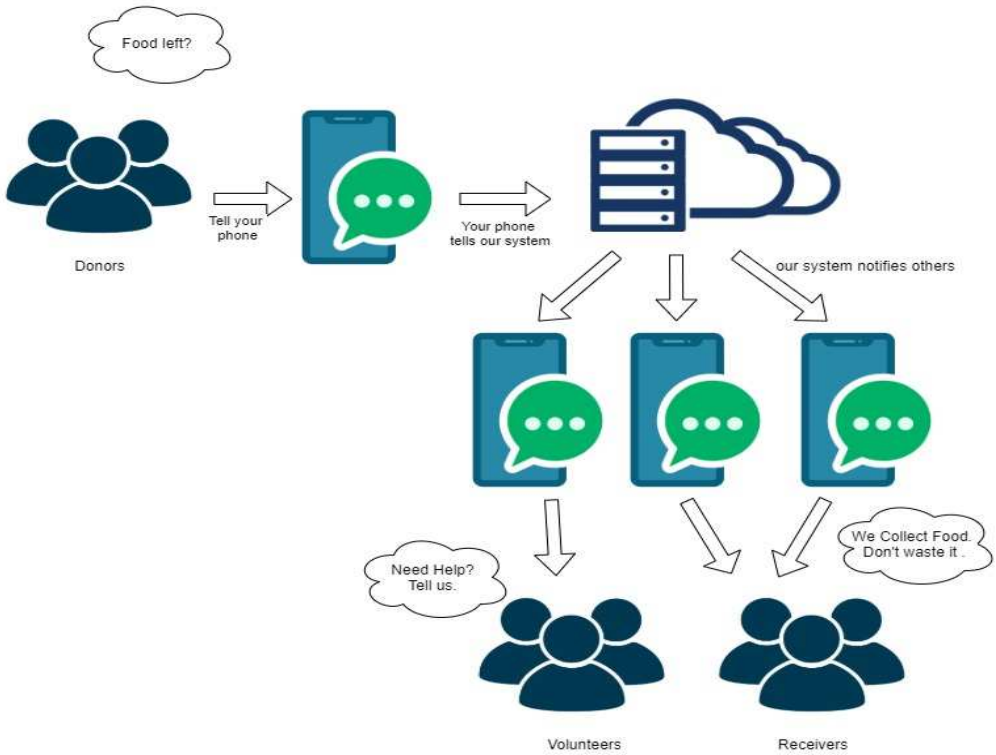

Fig. 1. Conceptual diagram of proposed solution

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IV. Proposed Work

A restaurant owner or any person can register himself as a helper (either donor or volunteer), while the organizations willing to get food donations from the application, will register themselves as a receiver.

If any helper wants to donate food to nearby organization, he can put up a donation notification, the hunger search algorithm filters all the organizations located within a pre-decided distance limit and that fits other donation criteria (e.g. Veg/Non-Veg, or number of people in the organization).

The donation notifications are sent to several organizations' caretakers (user or receiver), who are expected to be the point of contact of the respective organizations as scheduled by the algorithm. (more on this later). The notification vanishes as soon as receiver accepts the donation.

Either the donor or the receiver should be able to deliver food to the destination. If neither of them can deliver, receiver can put up a help-request which will notify all the users (helpers) in the vicinity. If any helper is interested to help, user can accept the pick-up request and deliver the food to the donation. This way it is possible to eliminate a single party maintaining all the logistics, giving control to the users to discuss and decide. After all it is believed that helping one-another is any person's responsibility.

Database Schema

The data is stored in unstructured format to enable easy distribution of continuously generated large number of donation notification among multiple machines. This helps in faster retrieval of needed information by decreasing traffic around a single machine, which means lesser documents to check in every machine.

Entities

- *User*

firebaseUid- a unique id that refers to user.

name, phone, email, area, city, state and pincode describe general user information.

Type - There are two types of users in the application, A person who helps or donates, and a person who takes care of an organization and receives donation notifications on behalf of the organization. I.e. Donor (Helper) and a Receiver

Organization - This is the unique string which carries a reference to an organization.

fcm_token - This is the unique string that identifies a mobile device, this string is useful for delivering messages to the mobile device of the user.

latitude and longitude are coordinates of the user's location. This is useful to calculate straight line distance and route distance between any two entities.

- *Organization*

Id - a unique id that refers to organization.

name, phone, area, city, state and pincode, description describe general information about the organization. latitude and longitude are coordinates of the organization.

isVeg - Boolean value to know whether organization prefers veg food only.

No_of_people - number of people who reside in the organization.

caretaker - This is the unique string which carries a reference to the user; the person to contact.

- *Donation*

id - a unique id that refers to notification.

isVeg - Boolean value to indicate whether the food is veg or non -veg.

No_of_meals - number of meals available for donation.

posted_at - time when the donation notification was posted.

Pick_up_latitude and pick_up_longitude are coordinates of the pickup location.

area, city, state and pincode, description describe general information. City, here is used to filter organizations inside the city.

posted_by, received_by and delivered_by are references to users who post, receive and deliver donations.

Self_drop indicates whether the donor will be able to deliver food by himself.

Status - integer to indicate donation status.

0 → *SEARCHING_FOR_RECEIVER*

1 → *CANCELLED*

2 → *REQUESTED_PICKUP*

3 → *DELIVERING*

4 → *EXECUTED*

last_modified - time when the donation notification status was modified.

Orgs - list of organizations for which the notification has to be delivered.

Token - the index of the organization in orgs which should be notified next.

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Likes - This feature is to maintain social engagement between users. References to users who have liked the executed donation.

Note - the names volunteer, donor and receiver refer to same entity user, different names in the context to be clearer about the different kinds of users.

Note - Notification refer to donation notification in this context.

Relations

- *One User → can post→ many Donation*

A user can donate as much as he can, so the user can add multiple donation notifications. but user can post only one donation notification at a time. One donation has to be complete before the user posts next one.

- *One User →takes care of→ one organization*

A person can receive donation notifications on behalf of an organization. This person will be the point of contact for the organization.

- *one Donation→can be sent to→many User*

A Donation Notification is supposed to notify multiple organizations, and hence their care-takers.

- *One User→ can accept→many Donation*

A user can accept one or more donations, but user can process only one at a time.

Hunger search Algorithm

Before proceeding to the actual algorithm, let us understand few pre-requisites listed below.

$$DISTANCE_LIMIT = 20$$

This will be the maximum radius in kilometers to find organizations looking for donations.

$$MEAL_DIFF_LIMIT = 5$$

This will be the maximum absolute difference between the number of people in organization and number of meals available for donation. A small difference is adjustable, because it is impossible to find the right fit.

A donation can be seen in 5 different stages, namely

$$0 \rightarrow SEARCHING_FOR_RECEIVER$$

User has posted a donation notification, he is waiting for the receiver to accept the donation.

$$1 \rightarrow CANCELLED$$

As name suggests the donation has been cancelled

$$2 \rightarrow REQUESTED_PICKUP$$

User has posted a donation notification, Receiver is found, but receiver has requested help to pick-up the donation.

$$3 \rightarrow DELIVERING$$

User has posted a donation notification, Receiver is found, Either Donor, Receiver or Volunteer is delivering the food to the destination.

$$4 \rightarrow EXECUTED$$

Donation successfully delivered to destination.

Using haversine formula [5] to calculate the great-circle distance between two geo-locations (latitude and longitude). An approximation of this value is used to filter organizations around the donor.

Haversine formula (la1, la2, ln1, ln2)

$$a = sin^2\left(\frac{\Delta la}{2}\right) + cos\,(la1) \times cos(la2) \times sin^2\left(\frac{\Delta ln}{2}\right) \quad (1)$$

$$c = 2 \times atan2(\sqrt{a} \times \sqrt{1-a}) \quad (2)$$

$$distance = R \times c \quad (3)$$

where,
$\Delta la = |la1 - la2|$
$\Delta ln = |ln1 - ln2|$
$la1 = source\ latitude$
$ln1 = source\ longitude$
$la2 = destination\ latitude$
$ln2 = destination\ longitude$
R is earth's radius (~6371km), latitude and longitudes are taken in radians

From the flowchart, it is seen that server is continuously monitoring the database, any change in the database triggers the server running on the cloud. Server is setup to act only on following circumstances.

- On creating a new donation notification.
- On update in existing donation notification.
- Every one minute.

Each scenario is discussed below.

- *On creating a new donation notification.*

step 1.

Add origin timestamp on donation creation

step 2.

Organizations = fetch all the organizations in the same city

$$i.e.\quad Oc = Nc \quad (4)$$

where

$Oc : Organization.city$
$Nc : notification.city$

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step 3.

Filter out locations off delivery limits

$$Od \geq DISTANCE_LIMIT \quad (5)$$

$where,$

$Od: Haversine_distance(Ola, Oln, Dla, Dln)$

$Od: Organization.distance$

$Ola: Organization.latitude$

$Oln: Organization.longitude$

$Dla: donation.pick_up_latitude$

$Dln: donation.pick_up_longitude$

step 4.

Filter out locations which doesn't match food-requirements

$$|On - Dn| > MEAL_DIFF_LIMIT \quad (6)$$

$where,$

$On: No\ of\ people\ in\ the\ orgnaization$

$Dn: No\ of\ meals\ available\ for\ donation$

step 5.

Order the remaining organizations in the non-decreasing order of the distance.

step 6.

Attach this list of organizations ids to the donation, organizations (orgs)

step 7.

Initialize the token to first organization in the list,

E.g., $token = 0$, meaning chance will be given to first organization

step 8.

Update organizations(orgs), token and posted_at of the donation

- *On updating existing donation notification*

step 1.

If $donation.status == 0$, do

If $i < dl$, do

Send notification to i^{th} organization's care_taker

$where$

dl: $number\ of\ shortlisted_organizations,$

i: $token\ number$

step 2.

If $donation.status == 1$, do

Send notification to donor

step 3.

If $donation.status == 2$, do

$helpers$=fetch all helpers in the same city i.e. $user.type == Donor$ and

$users.city == donation.city$

step 4.

Consider only helpers who lie in

$$Distance\ (d, h) < DISTANCE_LIMIT, \quad (7)$$

$where$

d: $donor$, h: $helper$

Alternatively, the ellipse method [6] can be used

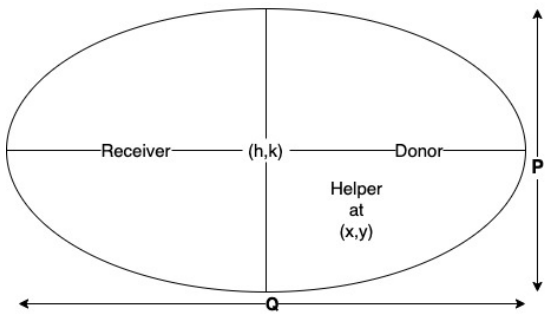

Fig. 2. elliptic boundary to find helpers in surroundings of donor and receiver

Find all the helpers who are at geo-location *(x,y)*, given that *(h,k)* is the midpoint equi-distant from both Receiver and Donor. *(h,k)* could be the origin of the imaginary ellipse with minor axis of length **P** and major axis of length **Q.** This approach can be more optimal in confining the total round-trip of the volunteer to certain limits as compared to using circle. But here it is needed to decide two parameters, P and Q.

$$\frac{(x-h)^2}{P^2} + \frac{(y-k)^2}{Q^2} <= 1 \quad (6)$$

- Send notifications to all of them, who pass the above decided criteria.
- Send notification to $donation.posted_by$ i.e. $donor$

  It is necessary to update donor that receiver has accepted request and requested for help

step 5.

If $donation.status == 3$, do

It is important to inform both donor and receiver that a volunteer has accepted the pick_up request.

- send notification to $donation.posted_by$,
- send notification to $donation.received_by.caretaker$,

step 6.

If $donation.status == 4$, do

When the order has been delivered successfully, inform donor, receiver and the volunteer

- send notification to $donation.posted_by$,
- send notification to $donation.delivered_by$

- *Every minute*

If a notified receiver does not respond within one minute, inform next receiver in the list.

Note: Receiver once notified, can view and respond to the donation notification even after that one minute, given that no other receiver has accepted it already.

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The food is most likely to be perishable and hence the donor cannot wait for long. So, algorithm cancels the donation by itself after a span of 60 minutes.

step 1.
$liveDonations$ = Fetch all the live notifications
i.e., $donation.status == 0$
step 2.
For every notification in $liveDonations$, do
If $timeDifference(dp, now) \geq 59\ minutes$, do
Update $donation.status = 1$
Continue
Otherwise do,
Update $dt = dt + 1$
i.e., inform the next organization in the list.

$where,$
$dp : donation.posted_at$
$dt : donation.token$

Note: It need not explicitly notify the receiver, on update function will take care of sending notifications to user, depending on the update.

Technologies used

- *Flutter*

Flutter is an open-source User Interface software development kit or a framework created by Google. Flutter is used to develop cross platform mobile application for Android and iOS.

- *Provider*

Provider is a facility in flutter framework to hold state information of the different entities interacting.

- *Dart*

Dart is a programming language used in front end development. The language is object-oriented, class-based and garbage-collected with C-style syntax.

- *Firebase*

Firebase is developed by google as well. It offers cloud services in the form of Platform as a Service (PAAS)[9]. Firebase provides number of features. But five features of the firebase are used primarily in the application.

- *Firestore Database*
  A simple No-SQL database to store unstructured collection of data documents.
- *Cloud Functions*
  Functions that are deployed on internet and gets triggered upon occurrence of certain event.
- *Firebase Authenication*
  A secure authentication system which provides variety of login and signup methods
- *Firebase Messaging*
  A service to send push notifications to users' devices.
- *Firebase Analytics*
  A service which keeps monitoring the activities and performance of the firebase project.

- *Location services*

Library containing functions to find out geo-locations of the device in the form of latitude and longitude.

- *Google Maps & Directions API*

Google service to calculate route-distances and display routes in the application.

V. Experimental results

Different use-cases that give insights about actual working of the algorithm:

Case 1:

Donor: put the donation notification
Receiver: Still not found
Self-drop: true
What will happen?
The donor will wait for maximum of 60 minutes, if no receiver accepts the notification. Every minute one new receiver will be notified in next 60 minutes, after that, notification self-expires.

Case 2:

Donor: put the donation notification
Receiver: Accepted the donation
Self-drop: true
Can receiver pickup: no
What will happen?
The donor will deliver the food to destination.

Case 3:

Donor: put the donation notification
Receiver: Accepted the donation
Self-drop: false
Can receiver pickup: yes
What will happen?
The receiver will go pickup food from the location.

Case 4:

Donor: put the donation notification
Receiver: Accepted the donation
Self-drop: false

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Can receiver pickup: no
What will happen?
Delivery Notification will be sent to the donor and other donors with details about donation, location and route.

Case 4.1:

Helper accepts the pick-up request: yes
Helper picks up donation and delivers to receiver. Receiver has to update the status once he receives the order.

Case 4.2:

Helper accepts the pick-up request: no
Receiver keeps waiting until no one responds, if no helper responds either donor or receiver are requested to contact each other with details provided and make necessary arrangements for delivery.

Demonstration of application

In figure 3.a and figure 3.b, Donor creates a donation notification by tapping on donate on homescreen, user provides required information and publishes it by clicking on submit. The user is expected to wait until he receives an update. The server monitoring the database will notify receivers as explained in the algorithm.

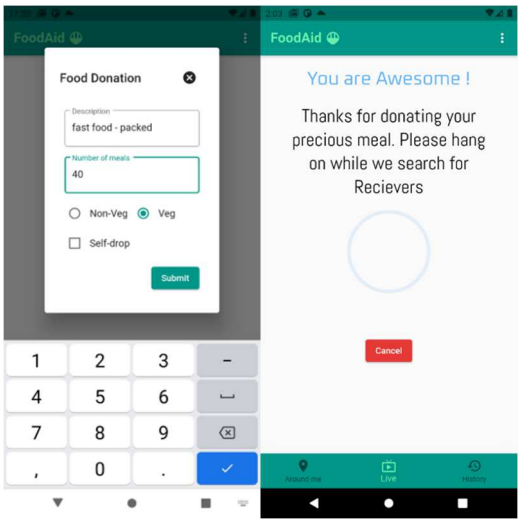

Fig. 3. a. when donor clicks on donate the form b. after submitting

Receiver is shown the donation notification on his live screen (figure 4), since the donor unselected self-drop, donor would not be delivering the donation. When the receiver clicks on accept, he/she is given two options, either to go pick up by themselves or request help from volunteers. Donor Is informed as well User clicks on 'request help', the server instantly notifies all the helpers in the vicinity.

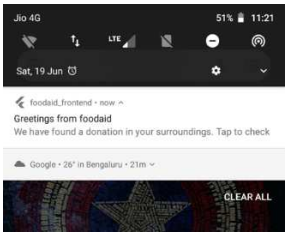

Fig. 4. push notification on receiver's device.

Figure 6.a and 6.b show that both donor (figure 6.a) and helper (figure 6.b) get notified when receiver requests for help. Either one of them can respond. In this case let us assume that volunteer (Manoj Kumar) accepts the request by clicking on help. The receiver and donor are then updated with the status of the donation.

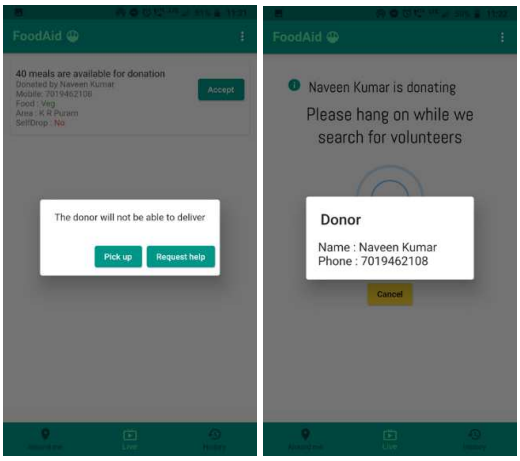

Fig. 5. a. Receiver clicking on tapping accept b. Receiver choosing request help option

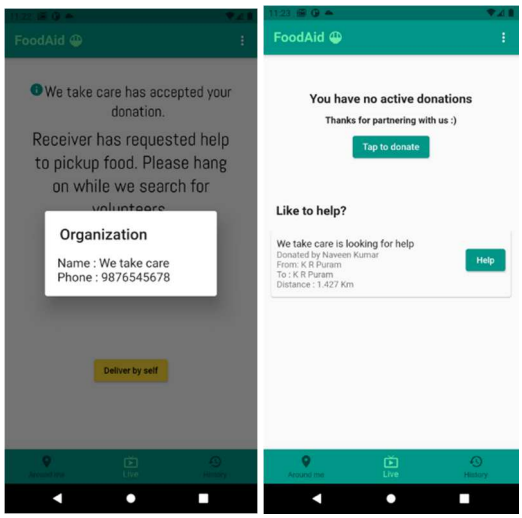

Fig. 6. updates on a. donor screen b. helper screen

The helper (as shown in figure 7.a) can accept multiple requests on his way. The receiver (as shown in figure 7.b) when they receive the donation successfully, is expected to notify us by tapping on

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delivered button. The donation as soon as it is delivered, is posted on 'around me' (figure 8 a) page, so others can appreciate their efforts. This page enables social engagement and serves as an encouragement to fellow donors.

Geo tagging to mark Hunger spots

Any helper can mark a location as a hunger-spot if there is a need (of help) in that particular location (on map). This will ideally help people to donate small scale house-hold left-overs to needy people near them.

The data collected can be used for variety of analysis tasks, for example, to figure out locations with most people living under poverty. This data could help redirect available services to the locations which needs it the most.

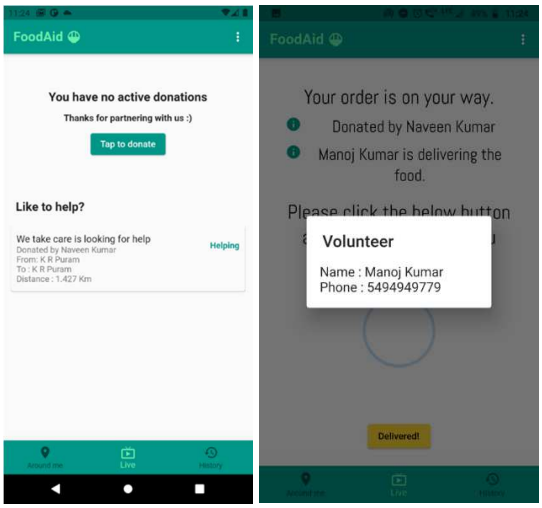

Fig. 7. a. Helper screen, b. Receiver screen

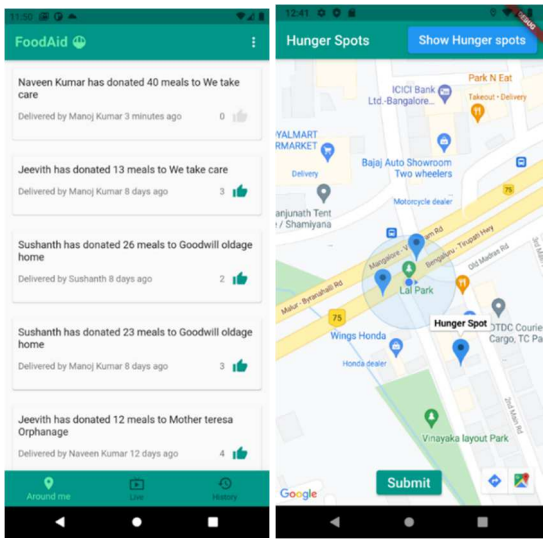

Fig. 8. a. around me page, b. hunger spots markings

TABLE 1 Comparison with existing Solutions

Model	Operation	Details
Restaurant owners offering free food through Food delivery applications	There could be food leftovers at restaurants, owners can put free food donation ads and the delivery agents collect food from assigned restaurant and delivers it to the intended receiver.	The Delivery agents get paid for their service as they do it for living. But the current application invites any person in the surrounding to volunteer for delivery.
Food donation Campaigns	There are several food donation campaigns happening across the country. Volunteers cook food and distribute to the needy people.	Such donations are kind and generous is definitely any human's duty, but they do not utilize leftover food, which is primary focus of the discussed application.
Organizations working for charity	Collecting funds from interested donors and redirecting funds to feed the poor.	Again, their focus is on feeding poor, while the current application's focus is on food leftover management.

VI. CONCLUSION AND FUTURE SCOPE

A restaurant owner or any person can register himself as a helper (either donor or volunteer), while the organizations willing to get food donations from the application, will register themselves as a receiver It is obvious that mobile phones have exerted a huge impact on society, and continue to do so to this day. The access to smart phones makes the lives of many easier and convenient.

Smart phones allow users to exchange any information they may need or want, whenever they want. Other applications, like Google Maps, function similarly by offering information within a moment's notice, such as travel directions and updates on traffic.

Mobile applications can be used to direct the food left over in both large scale and small scale to those who are actually in need. The users can register themselves as donors and receivers. The donors can post about the excess food they have, which reaches all the eligible receivers through the mobile application. The application runs on effective algorithm which manages the receivers based on their location and need of food.

After this sorting, filtering and notifying, the receivers can show their interest in receiving food wherein either donor or receiver can afford themselves for transportation. If both of them are not able to transport, another user who is willing to help can help them out. This approach of projecting the

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leftover food to those who are actually in need can make a big difference in food being wasted.

Flutter framework was used to develop the mobile application's front-end, and firebase to support the backend operations such as storing the data, authenticating users, deploying cloud functions. Cloud functions are essentially normal functions or methods which register their interest in data and gets triggered upon the change in data. This functions allow us to deliver notifications to qualified recipients as and when data is updated.

The algorithm shortlists organizations depending on the city, distance between participants and food requirements of the receiver. The application provides required information to enable direct contact between the participants.

However the proposed solution has few drawbacks, the organizations require us to manually verify the details provided. Despite of all the automation done in notifying users, being able to check whether food is in consumable state, is a primary responsibility of any user. And It is believed that donors are kind enough to assure that. After all the application's usage runs on trust.

The application's functionalities and performance has been tested under different scenarios. The system can be however improved or extended with new functionalities only upon receiving reviews from respective stakeholders. It is authors' hope that intended users make the best use of this product in their generous donations.

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